

Arithmetic Torsion Analysis for Knot 4_1

Note: For all examples below, the mapping used is 'a' as multiplicative ($\otimes_a$) and 'b' as additive ($\oplus_1$). For knot 4_1 , the Alexander polynomial is $\Delta(a) = a^2 - 3a + 1$. The cyclotomic polynomials referred to are $\Phi_1(a) = a - 1$ and $\Phi_2(a) = a + 1$. $p(a) = \nu(S_R)(0, a)$ and $q(a) = \nu(S_{R_{\text{rev}}})(0, a)$.

Table 1: Summary of Arithmetic Torsion Analysis for Knot 4_1 (Unified Mapping).

Relator(s) Used	$p(a)$	$q(a)$	Torsion $\tau(a) = \frac{p(a) - q(a)}{p(a) - q(a)}$	Torsion Factors (using $\Phi_n(a), \Delta(a)$)	Notes ($k_p, k_q, \sigma_{\text{eff}}$; Cyclot. Factors)
aaBAAbbbAB, abbbBAAB	$-\Delta(a)$	$-\frac{\Delta(a)}{a^2}$	$\frac{-\Delta(a)(a^2-1)}{a^2}$	$\frac{-\Delta(a)\Phi_1(a)\Phi_2(a)}{a^2}$	$k_p = 0, k_q = 2, \sigma_{\text{eff}} = -1$; Cyclot. $\Phi_1 \Phi_2$
aBAABabbb, bbbABaaBA, bbbBAABa	$-\frac{\Delta(a)}{a}$	$-\frac{\Delta(a)}{a}$	0	0	$k_p = 1, k_q = 1, p(a) = q(a)$; No cyclot. part

Arithmetic Torsion Analysis for Knot 6_2

Note: For all examples below, the mapping used is 'a' as multiplicative ($\otimes_a$) and 'b' as additive ($\oplus_1$). For knot 6_2 , the Alexander polynomial is $\Delta(a) = a^4 - 3a^3 + 3a^2 - 3a + 1$. The cyclotomic polynomials referred to are $\Phi_1(a) = a - 1$, $\Phi_2(a) = a + 1$, and $\Phi_4(a) = a^2 + 1$. $p(a) = \nu(S_R)(0, a)$ and $q(a) = \nu(S_{R_{\text{rev}}})(0, a)$.

Table 2: Summary of Arithmetic Torsion Analysis for Knot 6_2 (Unified Mapping, Grouped by Torsion Outcome).

Relator(s) Used	$p(a)$	$q(a)$	Torsion $\tau(a) = \frac{p(a) - q(a)}{p(a) - q(a)}$	Torsion Factors (using $\Phi_n(a), \Delta(a)$)	Notes ($k_p, k_q, \sigma_{\text{eff}}$; Cyclot. Factors)
aaBAAbbbAAbbbABaaBBB	$-\frac{\Delta(a)}{a^2}$	$-\frac{\Delta(a)}{a^2}$	0	0	$k_p = 2, k_q = 2, p(a) = q(a)$; No cyclot. part
aBAABBBAAAbbbbaabbb, bbbAAbbbABaaBBBaaBA, bbbBAABBBAAAbbbbaa	$-\frac{\Delta(a)}{a^3}$	$-\frac{\Delta(a)}{a}$	$\frac{\Delta(a)(a^2-1)}{a^3}$	$\frac{\Delta(a)\Phi_1(a)\Phi_2(a)}{a^3}$	$k_p = 3, k_q = 1, \sigma_{\text{eff}} = 1$; Cyclot. $\Phi_1 \Phi_2$
aabbbBAABBBAAAbbbb, bbbbaabbbBAABBBAAABa, bbbABaaBBBaaBabbbAA	$-\frac{\Delta(a)}{a}$	$-\frac{\Delta(a)}{a^3}$	$\frac{-\Delta(a)(a^2-1)}{a^3}$	$\frac{-\Delta(a)\Phi_1(a)\Phi_2(a)}{a^3}$	$k_p = 1, k_q = 3, \sigma_{\text{eff}} = -1$; Cyclot. $\Phi_1 \Phi_2$
aaBBBaaBAAbbaAbbaAB, abbaabbbBAABBBAAAB	$-\Delta(a)$	$-\frac{\Delta(a)}{a^4}$	$\frac{-\Delta(a)(a^4-1)}{a^4}$	$\frac{-\Delta(a)\Phi_1(a)\Phi_2(a)\Phi_4(a)}{a^4}$	$k_p = 0, k_q = 4, \sigma_{\text{eff}} = -1$; Cyclot. $\Phi_1 \Phi_2 \Phi_4$

Arithmetic Torsion Analysis for Knot 6_3

Note: For all examples below, the mapping used is 'a' as multiplicative ($\otimes_a$) and 'b' as additive ($\oplus_1$). For knot 6_3 , the Alexander polynomial is $\Delta(a) = a^4 - 3a^3 + 5a^2 - 3a + 1$. The cyclotomic polynomials referred to are $\Phi_1(a) = a - 1$, $\Phi_2(a) = a + 1$, and $\Phi_4(a) = a^2 + 1$. $p(a) = \nu(S_R)(0, a)$ and $q(a) = \nu(S_{R_{\text{rev}}})(0, a)$.

Table 3: Knot 6_3 Analysis: Type A Torsion $\tau(a) = \frac{\Delta(a)(a^4-1)}{a^4}$.

Relator(s) Used (Original Index)	$p(a)$	$q(a)$	Torsion $\tau(a)$	Torsion Factors	Notes
aabbaBAABabbaabABBAbABBAb (1), baabbbbaabABBAABabbaBAABBA (4), aBBabaBBabAAbbaBaaBabbAAb (6), aBBaaBabbABaaBBabAAbbaAAb (7), aabbbbaabABBAABabbaBAABBAb (8), baBBaaBabbABaaBBabAAbbaAA (17), baabbaBAABabbaabABBAABBA (27), baBBabaBBabAAbbaBaaBabbAA (30)	$\Delta(a)$	$\frac{\Delta(a)}{a^4}$	$\frac{\Delta(a)(a^4-1)}{a^4}$	$\frac{\Delta(a)\Phi_1\Phi_2\Phi_4}{a^4}$	$k_p = 0, k_q = 4, \sigma_{\text{eff}} = 1$; Cyclot. $\Phi_1 \Phi_2 \Phi_4$

Table 4: Knot 6₃ Analysis: Type B Torsion $\tau(a) = 0$.

Relator(s) Used (Original Index)	$p(a)$	$q(a)$	Torsion $\tau(a)$	Torsion Factors	Notes
aBAABBAbaabbbbaabABBAABabb (2), bABBAbaabbaBAABabbbaabABBA (9), aBBabAAbbABaaBabbAAbaBBab (11), aabABBAABabbaBAABBAbaabbb (16), bbaBAABBAbaabbbbaabABBAABa (18), bbAAbabBBabaBBabAAbbABaaBA (19), bbaBAABabbaabABBAabABBAbaa (20), bbABaaBBabAAbbbAAbabBBaaBA (21), bbbAAbabBBaaBabbABaaBBabAA (22), bbaabABBAabABBAbaabbaBAABa (23), baBBabAAbbABaaBabbAAbbaBBa (24), aBAABabbaabABBAabABBAbaabb (25), bbbaabABBAABabbaBAABBAbaa (31), aabABBAabABBAbaabbaBAABabb (32), bbABaaBabbAAbabBBabaBBabAA (34)	$\frac{\Delta(a)}{a^2}$	$\frac{\Delta(a)}{a^2}$	0	0	$k_p = 2, k_q = 2, p(a) = q(a)$; No cyclot. part

Table 5: Knot 6₃ Analysis: Type C Torsion $\tau(a) = \frac{\Delta(a)(a^2-1)}{a^3}$.

Relator(s) Used (Original Index)	$p(a)$	$q(a)$	Torsion $\tau(a)$	Torsion Factors	Notes
aaBAbbAAbabBBabaBBabAAbbAB (3), aaBBabAAbbbAAbabBBaaBabbAB (5), abbaabABBAabABBAbaabbaBAAB (10), abbaBAABBAbaabbbbaabABBAAB (12), abaBBabAAbbABaaBabbAAbbaBB (14), aaBAbbABaaBBabAAbbbAAbbaBB (29)	$\frac{\Delta(a)}{a}$	$\frac{\Delta(a)}{a^3}$	$\frac{\Delta(a)(a^2-1)}{a^3}$	$\frac{\Delta(a)\Phi_1\Phi_2}{a^3}$	$k_p = 1, k_q = 3, \sigma_{\text{eff}} = 1$; Cyclot. $\Phi_1\Phi_2$

Table 6: Knot 6₃ Analysis: Type D Torsion $\tau(a) = \frac{-\Delta(a)(a^4-1)}{a^4}$.

Relator(s) Used (Original Index)	$p(a)$	$q(a)$	Torsion $\tau(a)$	Torsion Factors	Notes
bABBAabABBAbaabbaBAABabbaa (13), bAAbbbAAbabBBaaBabbABaaBBa (26), bAAbbABaaBabbAAbabBBabaBBa (33), bABBAABabbaBAABBAbaabbbbaa (35)	$\frac{\Delta(a)}{a^4}$	$\Delta(a)$	$\frac{-\Delta(a)(a^4-1)}{a^4}$	$\frac{-\Delta(a)\Phi_1\Phi_2\Phi_4}{a^4}$	$k_p = 4, k_q = 0, \sigma_{\text{eff}} = -1$; Cyclot. $\Phi_1\Phi_2\Phi_4$

Table 7: Knot 6₃ Analysis: Type E Torsion $\tau(a) = \frac{-\Delta(a)(a^2-1)}{a^3}$.

Relator(s) Used (Original Index)	$p(a)$	$q(a)$	Torsion $\tau(a)$	Torsion Factors	Notes
abAAbbABaaBabbAAbabBBabaBB (15), abAAbbbAAbabBBaaBabbABaaBB (Index 28)	$\frac{\Delta(a)}{a^3}$	$\frac{\Delta(a)}{a}$	$\frac{-\Delta(a)(a^2-1)}{a^3}$	$\frac{-\Delta(a)\Phi_1\Phi_2}{a^3}$	$k_p = 3, k_q = 1, \sigma_{\text{eff}} = -1$; Cyclot. $\Phi_1\Phi_2$

Arithmetic Torsion Analysis for Knot 7_6

Note: For all examples below, the mapping used is 'a' as multiplicative ($\otimes_a$) and 'b' as additive ($\oplus_1$). For knot 7_6 , the Alexander polynomial is $\Delta(a) = a^4 - 5a^3 + 7a^2 - 5a + 1$. The cyclotomic polynomials referred to are $\Phi_1(a) = a - 1$, $\Phi_2(a) = a + 1$, and $\Phi_4(a) = a^2 + 1$. $p(a) = \nu(S_R)(0, a)$ and $q(a) = \nu(S_{R_{\text{rev}}})(0, a)$.

Table 8: Knot 7_6 Analysis (Unified Mapping): Type A Torsion $\tau(a) = 0$.

Relator(s) Used (Original Index)	$p(a)$	$q(a)$	Torsion $\tau(a)$	Torsion Factors	Notes
abABBBAbAABabbbbaBAbabABabbbbaAAB (2), abAAbabABbbABaaBAbABBBabABaaBAbbbAB (5), aaBAbbbABabAAbABabbbABaaBAbABBBabAB (6), abbbbaAABabABBBAbAABabbbbaBAbabAB (12), abABaaBAbbbABabAAbABbbABaaBAbABBB (21)	$\frac{\Delta(a)}{a^2}$	$-\frac{\Delta(a)}{a^2}$	0	0	$k_p = 2, k_q = 2, p(a) = q(a)$; No cyclot. part

Table 9: Knot 7_6 Analysis (Unified Mapping): Type B Torsion $\tau(a) = \frac{\Delta(a)(a^2-1)}{a^3}$.

Relator(s) Used (Original Index)	$p(a)$	$q(a)$	Torsion $\tau(a)$	Torsion Factors	Notes
aBAABabABBBAbAABabbbbaBAbabABabbb (3), bbbABABabABBBABaBAABabbbbaBAbabABa (14), bABBBAbAABabbbbaBAbabABabbbbaAABa (15), bAABabABbbABaaBAbABBBabABaaBAbbbABa (18), bbbABabAAbABbbABaaBAbABBBabABaaBA (19)	$\frac{\Delta(a)}{a^3}$	$-\frac{\Delta(a)}{a}$	$\frac{\Delta(a)(a^2-1)}{a^3}$	$\frac{\Delta(a)\Phi_1\Phi_2}{a^3}$	$k_p = 3, k_q = 1, \sigma_{\text{eff}} = 1$; Cyclot. $\Phi_1\Phi_2$

Table 10: Knot 7_6 Analysis (Unified Mapping): Type C Torsion $\tau(a) = \frac{-\Delta(a)(a^2-1)}{a^3}$.

Relator(s) Used (Original Index)	$p(a)$	$q(a)$	Torsion $\tau(a)$	Torsion Factors	Notes
aBBBabABaaBAbbbABabAAbABabbbABaaBAb (1), aBAbabABabbbbaAABabABBBABaBAABabbb (4), aabABabbbbaAABabABBBABaBAABabbbbaBAb (7), baBBBabABaaBAbbbABabAAbABabbbABaaBA (10), bbbABaaBAbABBBabABaaBAbbbABabAAbBA (11), bbbABAbabABabbbbaAABabABBBABaBAABa (13), baabABabbbbaAABabABBBABaBAABabbbbaBA (16), aBAbbbABaaBAbABBBabABaaBAbbbABabAAb (17), aBAABabbbbaBAbabABabbbbaAABabABBBAB (20), baBAbbbABaaBAbABBBabABaaBAbbbABabAA (22), baBAABabbbbaBAbabABabbbbaAABabABBBa (23)	$\frac{\Delta(a)}{a}$	$-\frac{\Delta(a)}{a^3}$	$\frac{-\Delta(a)(a^2-1)}{a^3}$	$\frac{-\Delta(a)\Phi_1\Phi_2}{a^3}$	$k_p = 1, k_q = 3, \sigma_{\text{eff}} = -1$; Cyclot. $\Phi_1\Phi_2$

Table 11: Knot 7_6 Analysis (Unified Mapping): Type D Torsion $\tau(a) = \frac{-\Delta(a)(a^4-1)}{a^4}$.

Relator(s) Used (Original Index)	$p(a)$	$q(a)$	Torsion $\tau(a)$	Torsion Factors	Notes
$\text{abbbabAbaabABabbbbaBAABabABBBAbabAAB-} \Delta(a)$ $(8), \text{aaBAbaBBBabABaaBAbbbABabAAbaBabbbAB}$ (9)		$-\frac{\Delta(a)}{a^4}$	$\frac{-\Delta(a)(a^4-1)}{a^4}$	$\frac{-\Delta(a)\Phi_1\Phi_2\Phi_4}{a^4}$	$k_p = 0, k_q = 4, \sigma_{\text{eff}} = -1;$ Cyclot. $\Phi_1\Phi_2\Phi_4$